SUPERPOSITION

A Quantum Computing Publication | Week 1 | Science Foundation Series

What Is Quantum Computing

And Why Is It Different?

"What if a computer did not think in 0s and 1s?"

Science Foundation Series | 2026

Topics: Qubits Superposition Entanglement Quantum Gates Bloch Sphere

Introduction

Every computer you have ever used, from the phone in your pocket to the servers running your favourite apps, operates on the same fundamental principle: information is broken into tiny switches called bits, each holding either a 0 or a 1. This binary logic has powered seven decades of remarkable progress.

But there is a class of problems so complex that even the fastest classical supercomputers would need longer than the age of the universe to solve them. Drug molecules with thousands of interacting atoms. Optimisation problems with billions of possible routes. Cryptographic keys built from enormous prime numbers. For these challenges, a fundamentally different kind of machine is needed.

THE CENTRAL QUESTION

What if a computer did not have to choose between 0 and 1? What if it could hold both values at once, exploring every possible answer simultaneously? That is the promise of quantum computing.

Quantum computing is not simply a faster version of what already exists. It operates according to the strange rules of quantum mechanics, the branch of physics that governs particles at the atomic and subatomic scale. This article explains the core concepts from the ground up, with no prior physics knowledge required.

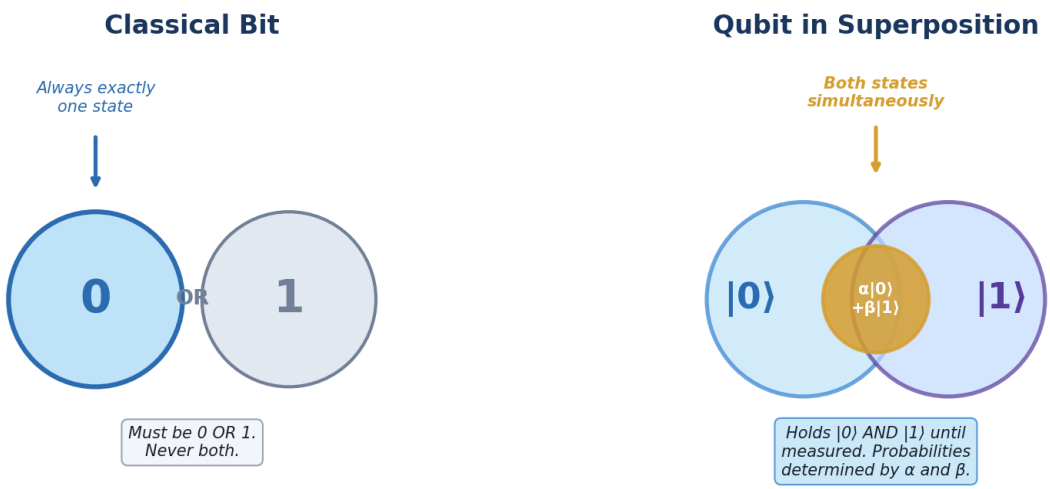
1. Classical Computing: The World of Bits

Before quantum computing can make sense, it helps to understand exactly what classical computers do and where they reach their limits.

How Classical Bits Work

A classical bit is the smallest unit of information in a traditional computer. It is a physical switch: a transistor that is either on (1) or off (0). Modern processors pack billions of these transistors onto a chip smaller than a fingernail. Every calculation, image, word, and video is ultimately represented as a long sequence of 0s and 1s.

This binary system is elegant and robust. Because each bit can only be 0 or 1, there is no ambiguity in the data, and errors are easy to detect. The downside is that classical computers process information one possibility at a time.



Classical Bit vs Qubit in Superposition

Figure 1: A classical bit must be either 0 or 1. A qubit in superposition holds both states simultaneously, described by the formula $\alpha|0\rangle + \beta|1\rangle$.

Where Classical Computers Struggle

For most everyday tasks, classical computers are perfectly adequate. But certain problems scale exponentially in difficulty as they grow in size. Consider a few examples:

- Finding the shortest route between 20 cities requires evaluating roughly 2 quintillion possible routes.
- Simulating a molecule of just 50 atoms requires more computational states than classical memory can represent.
- Breaking modern encryption keys relies on the fact that factoring large numbers classically would take millions of years.

SCALE OF THE PROBLEM

The Frontier supercomputer at Oak Ridge National Laboratory, among the most powerful classical machines ever built, can perform over a quintillion calculations per second. Yet there are molecular chemistry problems that would still defeat it, not because it runs slowly, but because the problem space itself grows too fast for any classical approach.

This is not a hardware limitation waiting to be solved by a faster chip. It is a mathematical ceiling built into how classical computation works. Quantum computing offers a genuinely different architectural approach to these specific problem types.

2. Enter the Qubit

A qubit, short for quantum bit, is the quantum equivalent of a classical bit. Like a classical bit, a qubit has two basis states, labelled |0> and |1> (read as "ket 0" and "ket 1" in the notation physicists use). Unlike a classical bit, however, a qubit can exist in a combination of both states at the same time, right up until the moment it is measured.

Physical Implementations

Qubits are not abstract concepts. They are real physical objects whose quantum states can be

controlled and read out. Several different technologies are currently used to build them:

Technology	How It Works	Who Uses It
Superconducting circuits	Tiny loops of superconducting metal cooled to near absolute zero, where current flows without resistance	Google, IBM, Rigetti
Trapped ions	Individual atoms suspended in electromagnetic fields and manipulated with laser pulses	IonQ, Quantinuum
Photons	Individual particles of light whose polarisation encodes qubit states	Xanadu, PsiQuantum
Topological qubits	Exotic quantum states engineered for inherent error resistance	Microsoft
Spin qubits	Quantum spin of electrons in semiconductor materials	Intel, academic labs

Each approach has trade-offs in stability, scalability, and operating temperature. Superconducting qubits, used by Google and IBM, must be cooled to temperatures colder than outer space: roughly 15 millikelvin, around 160 times colder than deep space.

3. Superposition: Being Two Things at Once

Superposition is the defining feature of quantum mechanics. It is the principle that a quantum system can exist in multiple states simultaneously, until it is observed or measured.

CORE DEFINITION

Superposition is a quantum system's ability to exist in multiple states at the same time. For a qubit, this means being in a combination of |0> and |1> simultaneously. The moment a measurement is made, the superposition collapses into one definite outcome.

The Spinning Coin Analogy

Imagine spinning a coin on a table. While it spins, it is neither heads nor tails. It is in a superposition of both possibilities. The moment it falls flat and stops, you see a definite result: heads or tails. A qubit behaves in a mathematically similar way. While it is evolving and interacting with other qubits, it holds both 0 and 1 as possibilities. The instant you measure it, the superposition collapses and you get a single answer.

This is not a metaphor for ignorance about which state the qubit "really" is in. Quantum mechanics is explicit: before measurement, the qubit genuinely does not have a single definite value. The superposition is a real physical condition, not just a gap in our knowledge.

Schrodinger's Cat

Physicist Erwin Schrodinger devised a famous thought experiment to illustrate how strange superposition becomes at larger scales. Imagine a cat in a sealed box with a device that has a 50% chance of releasing poison within one hour, triggered by a single radioactive atom. If the atom decays, the cat dies. If it does not, the cat lives.

According to quantum mechanics, before you open the box, the radioactive atom is in a superposition of "decayed" and "not decayed." This means the cat is simultaneously alive and dead, in a superposition of both outcomes, until the box is opened and the system is observed.

> *Schrodinger intended this as a reductio ad absurdum, pointing out how ridiculous quantum superposition seems when applied to everyday objects. But the underlying physics is experimentally confirmed. Particles really do exist in multiple states until measured. The cat scenario simply shows why quantum effects do not appear in macroscopic daily life: interaction with the environment causes decoherence, collapsing the superposition almost instantly at human scales.*

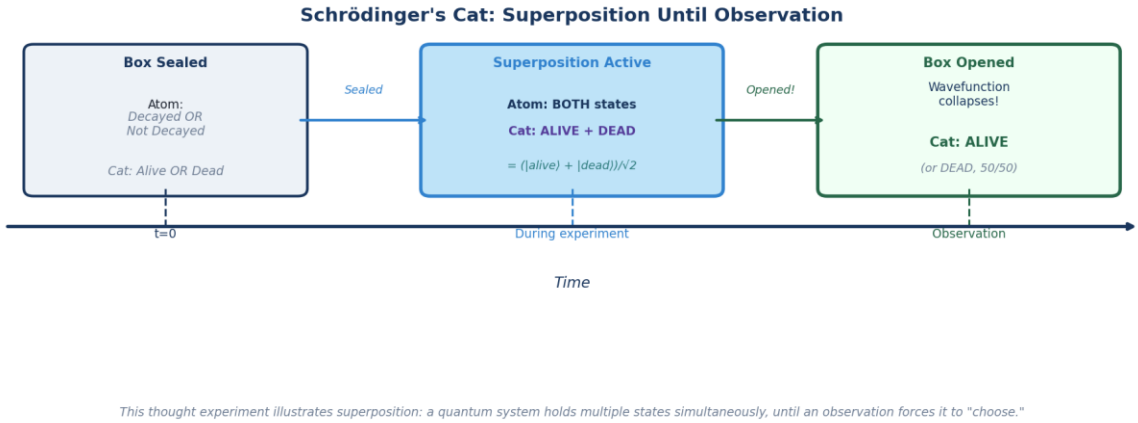


Figure 2: Schrodinger's Cat timeline. Before observation the system is in superposition. Opening the box collapses the quantum state to one definite outcome.

Why Superposition Is Powerful for Computing

The computational power of superposition comes from what it does when you have many qubits working together. A single qubit in superposition holds two possible states (0 and 1). Two qubits in superposition hold four possible states simultaneously. Three qubits hold eight. The number of simultaneous states doubles with every qubit added:

Number of Qubits	Simultaneous States	Classical Bits Needed to Represent
1 qubit	2 states (0 and 1)	1 bit
2 qubits	4 states	2 bits
10 qubits	1,024 states	1,024 bits
50 qubits	Over 1 quadrillion states	Over 1 quadrillion bits
100 qubits	1,267,650,600,228,229,401,496,703,205,376 states	More than all atoms in the observable universe
300 qubits	More states than atoms in the universe	Physically impossible classically

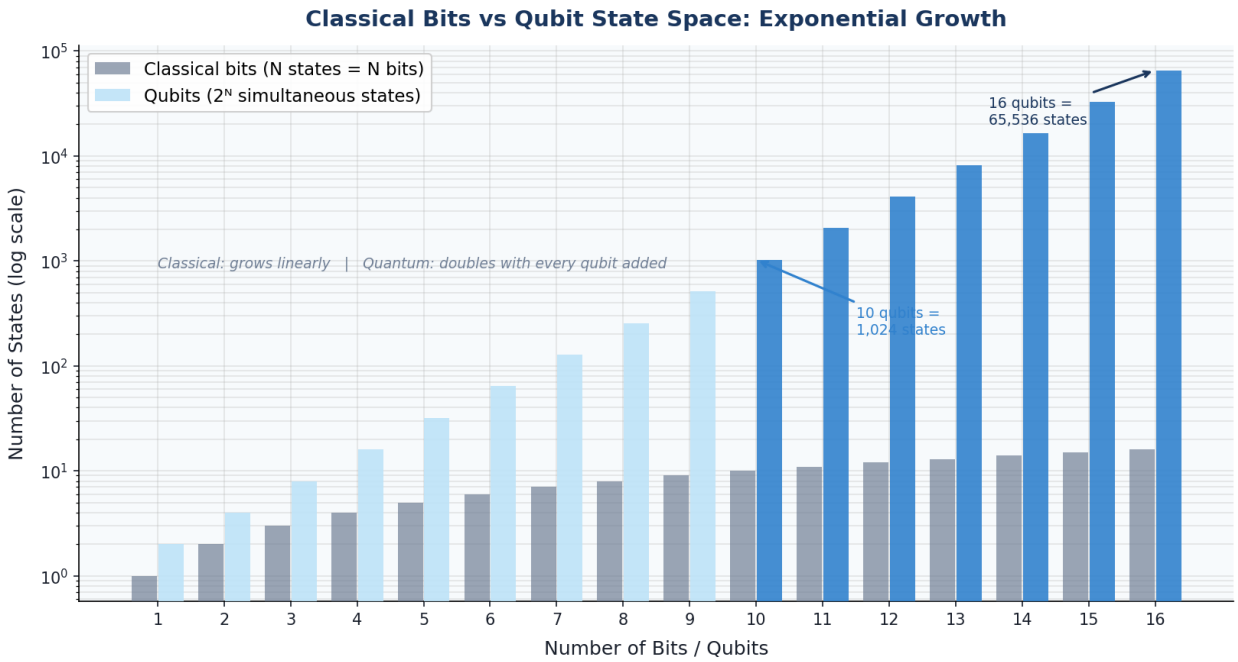


Figure 3: Classical bits grow linearly in state space. Qubits grow exponentially: every additional qubit doubles the number of simultaneous states.

This exponential scaling is what makes quantum computers so interesting for certain problem types. A quantum computer with 300 qubits in superposition is simultaneously holding more possible states than there are atoms in the observable universe. No classical storage medium can compete with that.

The critical caveat is that superposition alone is not enough. A quantum computer that simply puts all qubits into superposition and then measures them gives a random output with no more useful information than a coin flip. Superposition must be combined with quantum interference and entanglement, shaped by quantum algorithms, so that correct answers become more probable and wrong answers cancel out.

4. The Bloch Sphere: Visualising a Qubit

Physicists and quantum engineers use a geometric tool called the Bloch sphere to visualise the state of a single qubit. Understanding it helps build intuition for what superposition actually looks like mathematically.

How the Bloch Sphere Works

The Bloch sphere is an ordinary sphere with two special points:

- The North Pole represents the |0> state (the qubit will definitely measure as 0)
- The South Pole represents the |1> state (the qubit will definitely measure as 1)
- Every other point on the surface of the sphere represents a superposition of |0> and |1>

The position of a qubit state on the sphere is described by two angles: theta (the polar angle) and phi (the azimuthal angle). Theta determines the probability of measuring 0 or 1, while phi captures the relative phase between the two states, which matters for quantum interference.

Position on Bloch Sphere	What It Means	Theta Angle
North pole	|0> state: measuring always gives 0	0 degrees
South pole	|1> state: measuring always gives 1	180 degrees
Equator	Equal superposition: 50% chance of either	90 degrees
Northern hemisphere	More likely to measure 0	Between 0 and 90
Southern hemisphere	More likely to measure 1	Between 90 and 180

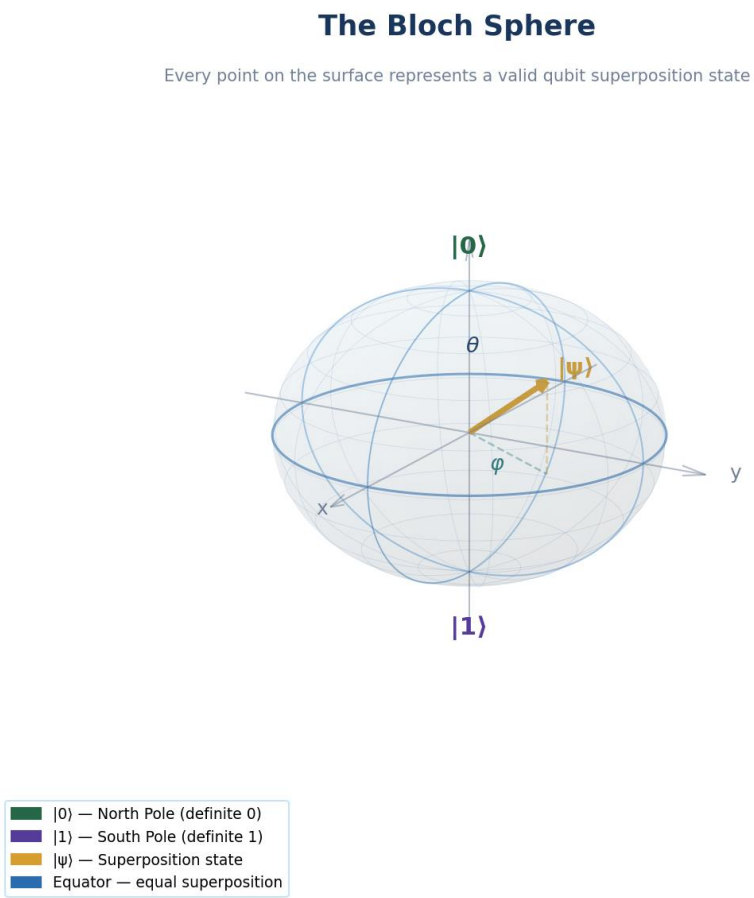


Figure 4: The Bloch Sphere. The amber arrow represents a qubit in superposition. θ (theta) sets measurement probabilities; φ (phi) captures the quantum phase.

WHY PHASE MATTERS

Two qubits on opposite sides of the Bloch sphere equator have the same measurement probabilities (50/50 either way) but completely different phases. This phase difference is invisible to a single measurement, but it drives quantum interference in algorithms, allowing correct answers to constructively amplify while wrong answers cancel out.

Try It Yourself: Interactive Bloch Sphere Simulators

Several free, browser based tools let you rotate a qubit state on the Bloch sphere interactively. These are excellent for building genuine intuition for superposition and quantum gates:

Resource	What You Can Do	Link
Bloch Sphere Simulator (kherb.io)	Drag a qubit state around the sphere, see angles update in real time	Open Simulator
IQM Academy Bloch Sphere	Interactive visualisation with gate controls and angle readouts	Open IQM
BlochSphereViz	Clean, minimal Bloch sphere with state vector display	Open Viz
Microsoft Quantum Superposition	Conceptual explainer paired with interactive demos	Open Microsoft
Quantum State Simulator (jonvet.com)	Experiment with multi-qubit states and gates	Open Simulator

5. Quantum Entanglement

Once two qubits interact in the right way, they can become entangled: a condition where their quantum states are correlated regardless of the physical distance between them. Measuring one qubit instantly determines what the other will show, even if the two particles are on opposite sides of the planet.

EINSTEIN'S "SPOOKY ACTION"

Albert Einstein famously called this "spooky action at a distance" because it seemed to violate common sense about locality. He believed it pointed to a flaw in quantum mechanics. Decades of experiments have since confirmed that entanglement is real. The 2022 Nobel Prize in Physics was awarded for precisely these experiments.

What Entanglement Is and Is Not

Entanglement is often misunderstood, and it is worth being precise about what it actually allows:

- Entangled qubits share correlated outcomes when measured. Measuring one qubit tells you what the other will show instantly.
- Entanglement does not allow faster-than-light communication. You cannot choose what outcome you get when you measure, so you cannot send information this way.
- Entanglement is a resource for quantum computation. It allows quantum computers to process correlations between qubits in ways that have no classical equivalent.

Entanglement in Computation

When qubits are entangled, operations on one affect the state of others. This interdependence allows quantum algorithms to process vast amounts of correlated information in a single step. Shor's algorithm for factoring large numbers, for instance, relies on entanglement to correlate the outcomes of many simultaneous calculations into a coherent answer.

The combination of superposition (multiple states at once) and entanglement (correlated states across qubits) is what gives quantum computers their characteristic exponential advantage over classical machines for specific problem types.

To explore entanglement further, the IEEE Spectrum guide to quantum entanglement provides a strong conceptual overview, and this SpinQ explainer connects it directly to computing applications.

6. Quantum Gates: The Logic of Quantum Circuits

In a classical computer, logic gates (AND, OR, NOT) manipulate bits. A quantum computer uses quantum gates to manipulate qubits, rotating their states on the Bloch sphere and creating entanglement between them. Every quantum algorithm is built from sequences of these gates.

Key Quantum Gates

Gate	Classical Equivalent	What It Does	Key Property
Hadamard (H)	None	Puts a qubit into an equal superposition of |0> and |1>	The primary gate for creating superposition
Pauli-X (X)	NOT gate	Flips |0> to |1> and |1> to |0>	Quantum NOT gate
Pauli-Z (Z)	None	Flips the phase of the |1> state without changing measurement probabilities	Phase manipulation
CNOT	XOR gate (partial)	Flips target qubit only if control qubit is |1>	Primary gate for creating entanglement
Toffoli (CCNOT)	AND gate	Flips target qubit only if both control qubits are |1>	Universal for reversible classical computation

THE HADAMARD GATE

The Hadamard gate (H) is one of the most important in quantum computing. Applied to a qubit in state |0>, it produces the superposition: (|0> + |1>) / sqrt(2). This means the qubit becomes 50% likely to measure as 0 and 50% likely as 1. Applied twice, it returns the qubit to |0>. It is the standard starting point for putting qubits into superposition before a quantum algorithm runs.

Unlike classical logic gates, all quantum gates are reversible. Given the output of a quantum gate, you can always reconstruct the input. This is a consequence of the time-reversibility of quantum mechanics and is a key difference from classical computation.

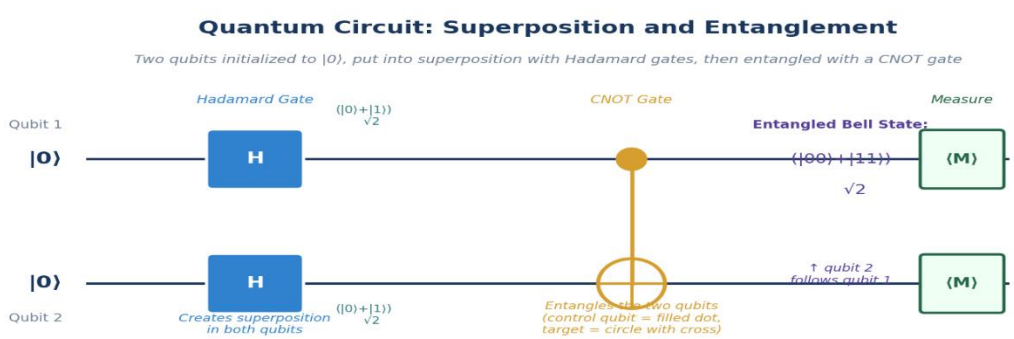


Figure 5: A quantum circuit. Two qubits start as |0⟩, pass through Hadamard (H) gates creating superposition, then interact through a CNOT gate which entangles them. Measuring either qubit instantly determines the other.

7. Why Classical Computers Struggle with These Problems

Quantum computing is not a general-purpose upgrade to classical computing. It offers dramatic advantages only for specific problem types, mostly those where the number of possible solutions grows exponentially with the size of the input.

The Maze Analogy

Imagine a classical computer solving a maze. It picks one path at a time and tries it. If the path reaches a dead end, it backtracks and tries another. In the worst case, it must try every possible path before finding the exit. A quantum computer, by contrast, can put the "position in the maze" into superposition, effectively exploring all paths at once. Quantum interference then amplifies paths that lead to the exit and suppresses dead ends.

This is a simplification, but it captures the essential difference. Quantum algorithms do not blindly try everything in parallel. They use interference to structure the exploration so that the right answer becomes disproportionately probable.

Two Landmark Quantum Algorithms

GROVER'S ALGORITHM (1996)

For an unstructured search through N items, a classical computer needs on average N/2 steps. Grover's algorithm needs only the square root of N steps. For one trillion items, that is the difference between 500 billion steps and about one million steps. It does not sound as dramatic as factoring, but this quadratic speedup applies to an enormous range of search and optimisation problems.

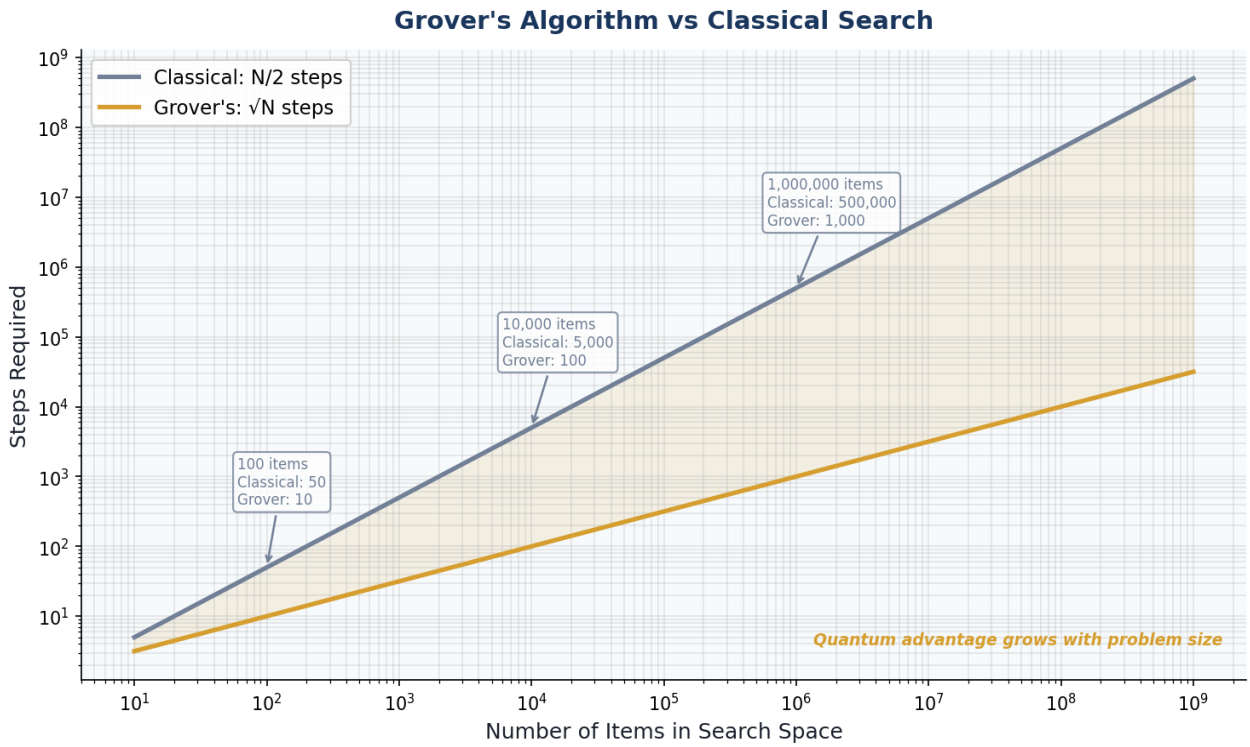


Figure 6: Grover's algorithm vs classical search on a log scale. The gap between the two lines grows rapidly, showing how quantum advantage increases with problem size.

SHOR'S ALGORITHM (1994)

Peter Shor's algorithm factors large numbers exponentially faster than any known classical method. The RSA encryption system used in banking and secure communications relies on the fact that multiplying two large prime numbers is easy, but factoring their product is practically impossible classically. Shor's algorithm would break this assumption on a large enough fault-tolerant quantum computer. (Present hardware is far too noisy and small to threaten real encryption today.)

Where Quantum Computing Does NOT Help

It is equally important to be clear about what quantum computers cannot do. They are not universally faster machines. For tasks like word processing, web browsing, video playback, or most database operations, a quantum computer offers no advantage and would actually be slower and more error-prone than a classical chip. Quantum speedups are real but highly specific.

8. Where Quantum Computing Can Make a Difference

The industries most likely to benefit from mature quantum computing are those dealing with exponentially complex problems where classical simulation reaches its limits.

Drug Discovery and Molecular Simulation

Designing a new drug molecule requires understanding how it interacts with thousands of atoms in a target protein. Classical computers can only approximate these interactions. A quantum computer can simulate quantum chemistry directly, exploring molecular energy landscapes with far greater fidelity. This could compress drug development timelines from years to months for certain molecule types.

Cryptography and Cybersecurity

Shor's algorithm poses a long-term threat to RSA encryption. Governments and standards bodies are already developing post-quantum cryptography standards, encryption methods that remain secure even against quantum computers. The US National Institute of Standards and Technology (NIST) finalised its first set of post-quantum standards in 2024.

Climate and Materials Science

Modelling how catalysts break down nitrogen for fertiliser production, or how solar cells convert light into electricity, requires quantum-level simulation. Even marginal improvements in catalyst efficiency or battery chemistry, enabled by quantum simulation, could have enormous real-world impact on energy and emissions.

Optimisation

Logistics, finance, and supply chain management all involve optimisation problems with vast numbers of variables. Finding the globally optimal shipping route, portfolio allocation, or manufacturing schedule is computationally intractable classically at large scales. Quantum algorithms offer potential speedups for these constrained optimisation problems.

Application Area	Classical Limitation	Quantum Potential
Drug discovery	Approximate molecular simulation only	Direct quantum chemistry simulation
Cryptography	Relies on factoring difficulty	Post-quantum standards in development
Climate modelling	Cannot model all quantum interactions	Higher-fidelity physical simulation
Materials science	Approximations for complex structures	Accurate prediction of material properties
Financial optimisation	Exponential solution space	Grover-based quadratic speedup
Machine learning	Classical linear algebra bottlenecks	Quantum matrix operations (speculative)

9. The State of Quantum Computing in 2025 and 2026

Quantum computing has moved from a largely theoretical discipline to an active engineering race. Several organisations have achieved milestones that were considered distant just a few years ago.

Google Willow: A Landmark Moment

GOOGLE WILLOW (DECEMBER 2024)

Google's 105-qubit Willow chip completed a benchmark computation in under five minutes that would take the fastest classical supercomputer an estimated 10 septillion years. More significantly, Willow demonstrated that error rates decrease as more qubits are added, a property called "below-threshold" error correction that had been theorised for decades but never demonstrated in hardware. This is widely regarded as one of the most important milestones in quantum computing history.

In October 2025, Google went further, demonstrating the Quantum Echoes algorithm on Willow: the first-ever verifiable quantum advantage on hardware, running approximately 13,000 times faster than the best classical supercomputer on a practical scientific task involving molecular structure.

Milestones Across the Field

Organisation	Development	Significance
Google	Willow chip (105 qubits, Dec 2024)	First below-threshold error correction; 10 septillion year benchmark
Google	Quantum Echoes algorithm (Oct 2025)	First verifiable quantum advantage on real hardware
IBM	Heron processor and modular architecture	Focus on gate fidelity and cloud quantum access via Qiskit
Microsoft	Topological qubit research	Pursuing inherently error-resistant qubits for long-term stability
IonQ	Trapped-ion quantum systems	Demonstrated practical advantages in molecular simulation tasks
NIST	Post-quantum cryptography standards (2024)	First official standards for quantum-safe encryption

Honest Assessment: Where Things Stand

The progress is real and accelerating. But it is worth being clear-eyed about the current state:

- Current quantum computers are still in the NISQ era (Noisy Intermediate-Scale Quantum). They have tens to hundreds of qubits but high error rates that limit the depth of circuits they can run.
- No quantum computer has yet broken real encryption or solved a commercially relevant optimisation problem that a classical computer could not handle with enough time.
- Fault-tolerant quantum computing, where error rates are low enough for indefinitely long calculations, likely requires millions of physical qubits per logical qubit and is probably at least a decade away from practical deployment.
- The gap between classical and quantum performance on real-world problems (as opposed to custom benchmarks) remains significant.

> *The responsible way to think about quantum computing in 2026 is this: the theoretical foundations are solid, the experimental progress is genuinely impressive, and the engineering challenges are hard but not insurmountable. The transformative applications are real, but they require hardware that does not yet exist at scale. Preparing now, learning the concepts, and watching the field develop, is the right approach.*

For more context on the current state of the field, see the NIST Quantum Information Science page and the MIT Technology Review quantum computing overview.

Glossary of Key Terms

Term	Category	Definition
Qubit	Fundamental Concept	The quantum equivalent of a classical bit. Unlike a bit, a qubit can exist in a superposition of 0 and 1 simultaneously.
Superposition	Theory	The quantum mechanical property allowing a system to exist in multiple states at once until measured.
Entanglement	Theory	A quantum correlation between two or more qubits such that measuring one instantly determines the state of the others, regardless of distance.
Quantum Gate	Algorithm	An operation that manipulates qubit states. Analogous to logic gates in classical computing but reversible.
Bloch Sphere	Fundamental Concept	A geometric representation of a single qubit's state as a point on the surface of a sphere. The north pole is |0>, the south pole is |1>.
Hadamard Gate	Algorithm	A quantum gate that places a qubit into equal superposition. The most common starting operation in quantum circuits.
Decoherence	Hardware	The loss of quantum superposition due to interaction with the environment. One of the biggest engineering challenges in quantum computing.
NISQ	Hardware	Noisy Intermediate-Scale Quantum. Describes current quantum devices with 50-1000 qubits but significant error rates.
Quantum Error Correction	Theory	Techniques that protect quantum information from errors by encoding logical qubits across many physical qubits.
Shor's Algorithm	Algorithm	A quantum algorithm for factoring large numbers exponentially faster than classical methods. Theoretically threatens current encryption.
Grover's Algorithm	Algorithm	A quantum search algorithm offering a quadratic speedup over classical unstructured search.
Wavefunction Collapse	Theory	The process by which a quantum system in superposition transitions to a single definite state upon measurement.

Further Reading and Interactive Resources

The following resources are selected for quality, accessibility, and depth. All links have been verified as of the publication of this article.

Interactive Learning Tools

- Bloch Sphere Simulator (kherb.io): https://bloch.kherb.io/
- IQM Academy Bloch Sphere Interactive: https://www.iqmacademy.com/play/bloch/
- BlochSphereViz: https://blochsphereviz.com/
- Quantum State Simulator: https://www.jonvet.com/tools/quantum-state-simulator
- Bits and Electrons Bloch Sphere Simulator: https://bits-and-electrons.github.io/bloch-sphere-simulator/

Educational Reading

- Caltech Science Exchange: What Is Quantum Superposition? scienceexchange.caltech.edu
- NIST: Quantum Computing Explained: nist.gov/quantum
- Microsoft Quantum: Superposition Explained: quantum.microsoft.com
- University of Maryland Quantum Atlas: quantumatlas.umd.edu
- SpinQ: What Is Quantum Superposition? spinquanta.com
- IEEE Spectrum: What Is Quantum Entanglement? spectrum.ieee.org

Recent News and Breakthroughs

- Google Willow Announcement (Dec 2024): blog.google/willow
- Google Quantum Echoes (Oct 2025): blog.google/quantum-echoes
- MIT Technology Review: What Is Quantum Computing? technologyreview.com
- QuantumZeitgeist: Superposition in Quantum Computing: quantumzeitgeist.com